TED (15)-4264 Reg. No. ………………………

(REVISION – 2015) Signature ……………………..

FOURTH SEMESTER DIPLOMA EXAMINATION IN INFORMATION TECHNOLOGY

MARCH, 2017

FOUNDATIONS OF COMPUTER HARDWARE

Model Question Paper

[*Time : 3 hours*]

(Maximum marks : 100)

PART – A

I Answer the following questions in one or two sentences. Each question carries 2 marks.

1. Define weighted code?
2. What is decoder?
3. Name any two application of flip flop
4. List any two application of Shift register
5. What is a flag register ? (5 x 2 = 10)

PART – B

II Answer *any five* of the following. Each question carries 6 marks.

1. Write the features of hexadecimal number system. How to convert it to decimal and binary.
2. Prove that (AB’(C+BD)+A’B’)C=B’C
3. Draw and explain the working of Full adder circuit ?
4. Explain the working of 4x1 Multiplexer
5. Draw the logic diagram and truth table of SR flip flop?
6. Draw the logic diagram and explain Ring counter?
7. Explain the instruction set of 8086 ? (5x 6 = 30)

[PTO]

PART – C

(Answer one full question from each unit. Each questions carries 15 marks.)

Unit – I

III a)Convert the following

(i)Octal number 234 to hexadecimal (ii)Hexadecimal EB3 to decimal

(iii)Decimal 32.16 to binary (iv) Decimal 57 to decimal 10 marks

b) State Demorgan's theorems 5 marks

OR

IV (a) Simplify using Karnaugh map F(A,B,C,D)=∑(5,6,7,9,10,11,13,15) 9 marks

(b)Write short note on Alphanumeric codes 6 marks

Unit – II

V (a) Explain the working of a serial adder ? 8 marks

(b) Explain 1x4 DEMUX with logic diagram and truth table? 7 marks

OR

VI (a) Design and implement a full subtractor 15 marks

Unit – III

VII (a) Compare Synchronous and Asynchronous counters 8marks

(b) Explain the working of SR flip flop 7marks

OR

VIII (a) Design 4-bit ripple counter and explain its working ? 15marks

Unit – IV

IX (a) Explain the working of serial in serial out shift register 9marks

(b) Classify the interrupts of 8086 6marks

OR

X (a) Explain the architecture of 8086 with block diagram 15marks